NeST-Com: Fixed-Budget Next-Scale Token Communication

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Abstract

Next-scale residual tokens provide a coarse-to-fine image description, but whole-scale endpoints can underfill fixed digital frames. NeST-Com fits independently terminated, entropy-coded within-scale prefixes and uses received tokens for residual reconstruction and finer-scale conditioning. With shared model components, PSNR improves over whole-scale transmission by 0.32–0.70 dB at 1024 symbols and 0.39–0.79 dB at 512 symbols. Source-bit utilization rises from 77.70% to 95.21% at 1024 symbols and 10 dB. At 512 symbols, NeST-Com exceeds a fixed-frame Ada-TokenCom adaptation by 1.74–4.18 dB PSNR; Ada achieves higher CLIP-I throughout and lower DreamSim at 19 dB.

Index Terms

Image communication, visual autoregressive models, fixed symbol budget, entropy coding.

I. Introduction

When each image has a fixed channel-symbol budget, the transmitter must decide which source information to retain, and the receiver must reconstruct the image from that description. Feasibility depends on the image-dependent source-code length and the protection carried within the frame. A generative prior can complete untransmitted information, while the delivered representation provides source-image constraints.

Digital coding with error correction and learned joint source–channel coding (JSCC) provide different solutions [1], [2]; fitting progressive descriptions to channel resources is a classical problem [3]. Ada-TokenCom combines prefix entropy coding, tail generation, and prefix/MCS selection under an average resource constraint [4]. TONIC studies task-oriented token communication [5], while ARPC constructs progressive descriptions with an autoregressive prior [6]. We adopt the next-scale representation of visual autoregressive modeling (VAR) [7]. Complete coarse maps provide a low-resolution description of the full latent grid; subsequent maps add residual information and are predicted from earlier scales. This organization supports reconstruction from a coarse description when only a limited prefix can be delivered.

Coarse-to-fine refinement does not make whole-scale boundaries align with digital-frame capacity. Adding a complete scale requires a large token increment, and the independently terminated entropy-code length of a given endpoint varies across images. After one complete scale fits, the remaining capacity may be insufficient for the next. Within-scale endpoints retain source tokens from that boundary map. The next-scale factorization lets both ends construct matching coding probabilities from complete coarse maps. To turn the additional received information into reconstruction gains, the boundary tokens must enter residual accumulation and the conditions used to predict finer scales.

NeST-Com combines within-scale endpoints, independently terminated arithmetic streams, actual-length fitting, and prefix-constrained reconstruction in a fixed-budget digital system. It uses VAR's known-token overwrite operation [7] to merge received boundary positions with predictions and update both the residual latent and finer-scale context. With shared model components, PSNR gains over whole-scale transmission are 0.32–0.70 dB at $N = 1024$ and 0.39–0.79 dB at $N = 512$; utilization rises from 77.70% to 95.21% at $N = 1024$, 10 dB. Controlled endpoint and receiver interventions examine these design gains. Six-system N1024 comparisons

and independently calibrated N512 results show the source-reconstruction performance and perceptual tradeoffs of the complete systems.

II. Proposed NeST-Com Framework

A. Next-Scale Representation and Digital Frame

We transmit $x \in [0,1]^{256\times256\times3}$ over complex AWGN,

$$y = s + n, \quad n \sim \mathcal{CN}(0, \sigma^2 I_N), \quad \mathrm{SNR} = 10\log_{10}(E_s/\sigma^2), \tag{1}$$

using N complex symbols per image and nominal $E_s = 2$. TX selects a frozen policy for the channel SNR; RX uses the true noise variance for LLRs. Digital constellations have average energy two without per-frame renormalization. Continuous JSCC data are normalized per image to average energy two, with required Swin side information paid within N. A 12-bit profile ID, CRC16, and terminated rate-1/2 convolutional code occupy 68 rotated-BPSK symbols $\{\pm(1+j)\}$. The remaining $N-68$ symbols carry the LDPC-coded body. The paid profile identifies the actual endpoint and MCS. Our main budget is $N = 1024$; the reduced-budget experiment uses $N = 512$ without increasing E_s.

The frozen VAR tokenizer produces ten residual token maps q_s, with side lengths $(1, 2, 3, 4, 5, 6, 8, 10, 13, 16)$ and a 4096-entry codebook [7]. Let L_s be the map size and C_s its codebook lookup, resizing, and residual transform. Complete coarse maps span the whole latent grid after resizing; the sum $f_q = \sum_s C_s(q_s)$ accumulates their residual refinements. Image decoder D_c is adapted on 20 000 ImageNet training images using continuous, fully quantized, and prefix-completion-interpolated latents, then fixed. NeST-Com uses null-class conditioning. Its frozen next-scale prior factorizes as

$$p_\theta(q_s \mid q_{<s}) = \prod_{i=1}^{L_s} p_\theta(q_{s,i} \mid q_{<s}), \tag{2}$$

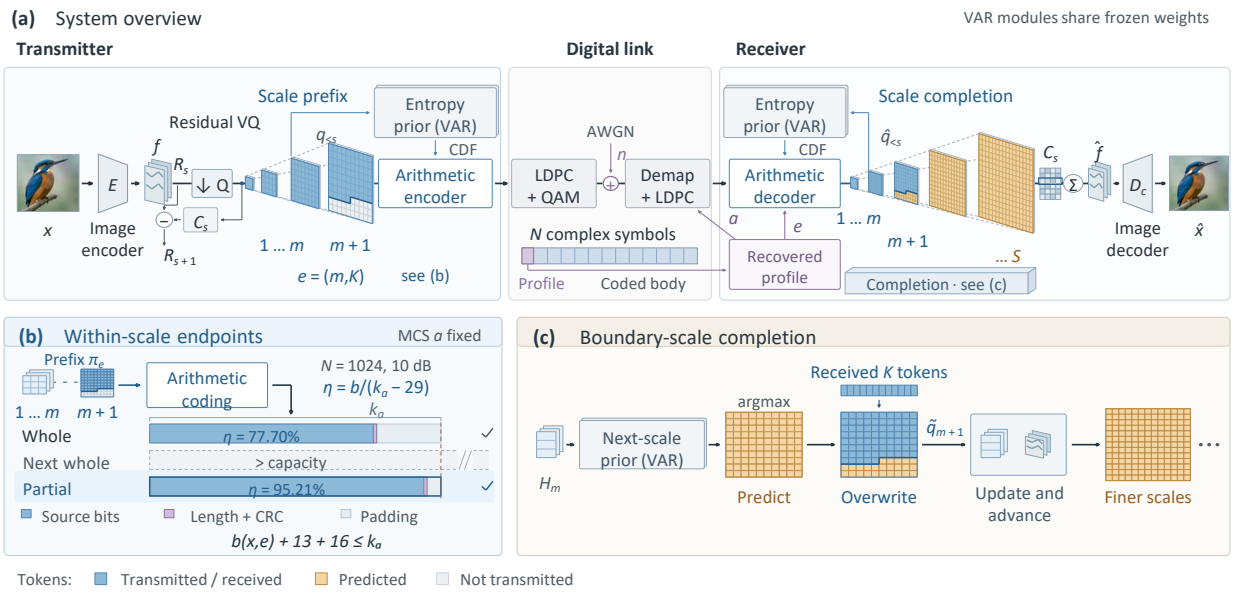


Fig. 1. NeST-Com. (a) The paid profile specifies endpoint e and MCS a; TX/RX construct CDFs from the shared frozen VAR and their known coarse tokens. (b) Independently terminated prefixes, a 13-bit length field, and CRC16 fit k_a. Bars show common500 entropy-stream capacity utilization at $N = 1024$, 10 dB (WHOLE 77.70%; PARTIAL 95.21%). (c) Received positions overwrite the predicted boundary map before residual accumulation and advancement to finer scales. H_m denotes the prior state after scale m.

providing local cumulative distribution functions (CDFs) for arithmetic coding [8] at both ends of Fig. 1.

B. Within-Scale Endpoints and Actual-Length Fitting

An endpoint $e = (m, K)$ retains complete coarse maps and K raster-ordered positions from the next map:

$$\pi_e = (q_1, \ldots, q_m, q_{m+1,1:K}). \tag{3}$$

WHOLE permits $K = 0$ for $m = 4, \ldots, 10$. PARTIAL adds $K \in \{\lfloor L_{m+1}/4 \rfloor, \lfloor L_{m+1}/2 \rfloor, \lfloor 3L_{m+1}/4 \rfloor\}$ for $m = 4, \ldots, 9$, giving 25 endpoints in total. $(10, 0)$ is the full representation.

For MCS $a = (M, r)$, the coded body has $n_a = (N - 68) \log_2 M$ bits and information length $k_a = \lfloor r n_a \rfloor$. With actual terminated arithmetic length $b(x, e)$,

$$b(x, e) + 29 \leq k_a, \qquad \eta(x, e) = \frac{b(x, e)}{k_a - 29}. \tag{4}$$

The endpoint determines which source tokens are retained, while its actual terminated length determines whether they fit the frame. The 29 bits comprise the length field and CRC16; unused information positions are zero-padded before 5G NR LDPC coding, rate matching [9], and QAM. Thus η measures entropy-stream capacity utilization.

Offline calibration selects a target endpoint and MCS for each budget/SNR. A 100-image pilot uses one channel realization per image; the finalists are evaluated on all 1000 calibration images with three realizations per image. Selection maximizes source/reconstruction DINOv2 cosine [10], averaged over realizations first and then equally over sources, including gray failures. Exact ties use a fixed candidate order. At $N = 1024$, target screening starts at scales 7–9 and includes full-scale $(10, 0)$ candidates in final evaluation. At $N = 512$, WHOLE/PARTIAL screen all 7/25 targets against the same 12 MCSs, $\log_2 M \in \{2, 4, 6\}$ and $r \in \{1/2, 2/3, 3/4, 5/6\}$. Each method/SNR advances the top three plus a maximum-endpoint QPSK rate-1/2 anchor. Ada uses the same N512 MCS pool with longest-feasible fitting up to 256 tokens. The full candidate and selection rules are provided with the results.

Online fitting fixes the selected MCS and searches legal endpoints no longer than the target, in decreasing token count down to $(4, 0)$. TX sends the first independently terminated stream satisfying (4); the paid profile identifies the actual endpoint. Arithmetic streams are never cut from longer streams. If no endpoint fits, the N512 protocol sends a paid NULL profile and short NULL payload through the complete frame, retaining a gray reconstruction in the denominator.

The N1024 implementation instead raises an error; no evaluated N1024 source encountered this case.

C. Prefix-Constrained Reconstruction

Invalid profile CRC/catalogue, body CRC, declared length, known padding, or arithmetic parsing produces a fixed gray image; so does a valid N512 NULL reception. Otherwise, RX decodes the prefix using CDFs rebuilt from received complete coarse maps. It predicts the boundary map by VAR argmax conditioned on those maps, then applies known-token overwrite [7]:

$$\tilde{q}_{m+1,i} = \begin{cases} \hat{q}_{m+1,i}, & i \le K, \\ q^{\mathrm{pred}}_{m+1,i}, & i > K. \end{cases} \tag{5}$$

Unreceived boundary positions retain their coarse-conditioned prediction. Received boundary tokens serve two roles: they correct the residual representation and, through the merged boundary map, condition the prediction of finer scales. Complete received maps remain unchanged; subsequent maps are generated from this updated context. Reconstruction is

$$\hat{x} = D_c\left(\sum_{s=1}^{10} C_s(\tilde{q}_s)\right). \tag{6}$$

At the final scale, overwrite changes the residual contribution without any subsequent propagation.

III. Experimental Results

A. Experimental Setup

We use common500 ImageNet validation images [11], a disjoint 1000-image calibration set, and the 24 Kodak images [12], with fixed 256×256 preprocessing. The six-system main experiment uses $N = 1024$; the reduced-budget study compares NeST, WHOLE, and Ada at $N = 512$. Both use $E_s = 2$ and 4/10/19 dB. ImageNet-calibrated policies remain fixed on Kodak.

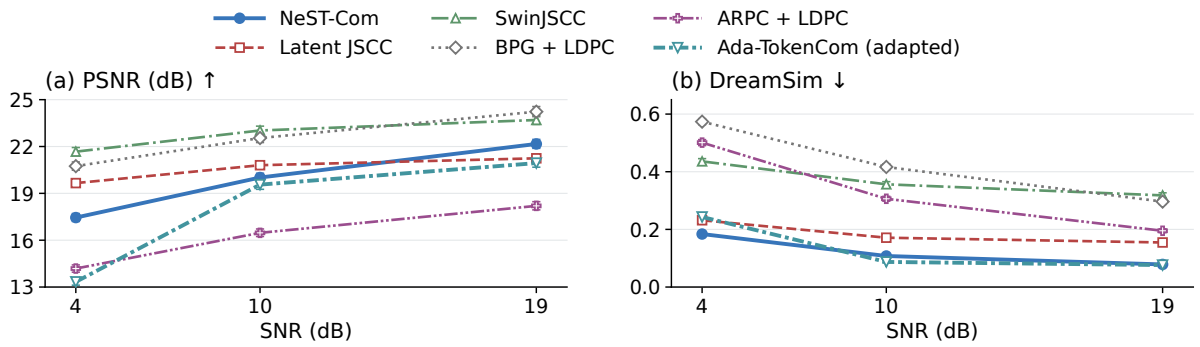


Fig. 2. Six-system performance on common500 at $N = 1024$. Points and bars are source means and pointwise 95% source-bootstrap CIs after averaging three channel realizations. Ada denotes the predicted-class, fixed-frame adaptation. Higher PSNR and lower DreamSim are favorable.

The six systems are NeST-Com, Latent JSCC, adapted SwinJSCC [2], adaptive BPG [13] + LDPC, adapted ARPC [6] + LDPC, and Ada-TokenCom adapted to a fixed per-image frame. Our convolutional Latent JSCC maps tokenizer latents to symbols and shares D_c; its losses combine image MSE, LPIPS, and normalized-latent consistency. Latent JSCC and SwinJSCC use the same 20 000 training images and final 100k/200k-update checkpoints. SwinJSCC was still improving at its training budget cap. All required signaling is charged; ARPC's caption is paid. Ada retains LlamaGen-L/VQ16, prefix arithmetic coding, and conditional tail sampling. A fixed ConvNeXt predicts the class, whose 10-bit ID is transmitted, giving $b+39 \leq k_a$. Calibration chooses its MCS; online fitting searches the predeclared 1–256-token endpoints. This changes Ada's average-resource formulation to our fixed-frame task.

Metrics are PSNR, LPIPS-Alex [14], DreamSim [15], and CLIP-I, the CLIP ViT-L/14 image–image cosine [16]. DreamSim/CLIP use frozen FP32, 224×224 preprocessing. We average three channel realizations per source, retaining gray failures, then compute equal-source means. Intervals are 95% source-bootstrap CIs; differences are directly source-paired. Pairing does not

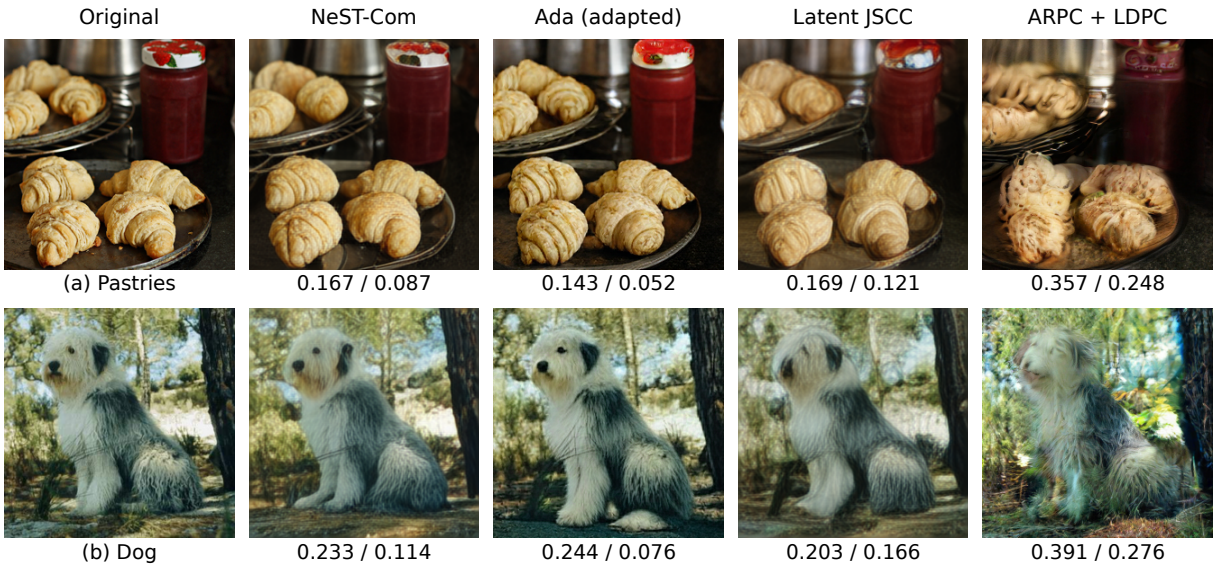


Fig. 3. Selected methods on two fixed development examples at $N = 1024$ and 10 dB: (a) pastries and (b) a dog. Values below each reconstruction are LPIPS-Alex↓ / DreamSim↓ for that displayed realization.

require identical noise realizations across methods. Generative seeds and decoding settings are fixed. N512 methods share source/repeat-specific AWGN; the historical N1024 systems retain their original noise realizations. At $N = 1024$, no rejected or failed reconstructions occurred in the 1500 trials per method at each matched SNR on common500. Continuous Latent JSCC has no discrete CRC rejection rule. Transmitter fallback is counted separately.

B. Effect of Within-Scale Endpoints

Table I(A) compares WHOLE/PARTIAL with shared tokenizer, prior, entropy coder, and D_c. Their target/MCS choices are calibrated separately and select the same MCS at each SNR; fallback follows the fixed rule. On common500, PSNR gains are 0.32–0.70 dB. At 10 dB, utilization rises from 77.70% to 95.21% and DreamSim decreases by 0.0240. Kodak gains 0.674 dB and reduces DreamSim by 0.0164 at 10 dB; its 19 dB DreamSim CI includes zero.

TABLE I

WITHIN-SCALE ENDPOINT EVIDENCE. (A) MAIN FAMILY-SPECIFIC POLICIES; $\Delta = \mathrm{PARTIAL} - \mathrm{WHOLE}$ WITH SOURCE-PAIRED 95% CIS. (B) CONTROLLED 100-SOURCE DIAGNOSTIC AT 10 DB.

(A) Main policies: utilization and paired PSNR gain

SNR	Utilization W→P (%)	ΔPSNR (dB) [95% CI]
common500		
4	73.81 → 93.08	$+0.616\ [0.564, 0.673]$
10	77.70 → 95.21	$+0.701\ [0.637, 0.766]$
19	85.24 → 94.73	$+0.319\ [0.290, 0.349]$
Kodak24		
4	75.69 → 93.43	$+0.396\ [0.236, 0.543]$
10	78.11 → 94.99	$+0.674\ [0.408, 0.958]$
19	85.69 → 93.71	$+0.228\ [0.140, 0.324]$

(B) Fixed model, MCS, maximum endpoint and fitting

Grid (candidates)	Utilization (%)	PSNR (dB)
Whole (7)	78.15	18.952
Half (13)	89.21	19.396
Quarter (25)	95.38	19.612
Eighth (49)	97.39	19.735

These results connect additional legal endpoints and calibrated fitting to better use of a fixed body capacity.

Block B isolates granularity on 100 existing sources, holding model, MCS, maximum endpoint, and longest-feasible fitting fixed. WHOLE, half-, quarter-, and eighth-scale grids have 7/13/25/49 candidates. At 10 dB, QUARTER minus WHOLE gains $0.660\,[0.515, 0.814]$ dB, while EIGHTH

TABLE II

PERCEPTUAL DISTANCE AND IMAGE-FEATURE SIMILARITY ON COMMON500. EACH CELL IS LPIPS-ALEX ↓ / CLIP-I ↑.

Method	4 dB	10 dB	19 dB
NeST-Com	0.263 / 0.821	0.171 / 0.873	0.118 / 0.900
Latent JSCC	0.219 / 0.741	0.173 / 0.789	0.158 / 0.804
SwinJSCC	0.530 / 0.627	0.469 / 0.669	0.440 / 0.697
BPG + LDPC	0.725 / 0.586	0.589 / 0.635	0.458 / 0.684
ARPC + LDPC	0.512 / 0.644	0.341 / 0.713	0.250 / 0.788
Ada (adapted)	0.395 / 0.828	0.167 / 0.884	0.138 / 0.892

adds $0.123\,[0.079, 0.169]$ dB over QUARTER. The denser grid also yields small PSNR/LPIPS gains at other SNRs; the main quarter grid remains the predefined configuration.

A receiver intervention uses the same 100 sources and actual received prefixes, without new endpoint selection or channel trials. A0 ignores received boundary positions and follows the prior path. B0 adds their correction only to the final residual latent, leaving finer-scale context on the A0 path. C0 updates both latent and context. All three use the same device and numerical precision. At 10 dB, B0$-$A0 and C0$-$B0 yield 0.220 and 0.440 dB mean PSNR differences. C0$-$B0 is zero at 19 dB because the boundary is the last scale.

C. Comparison With Existing Methods

Fig. 2 and Table II show a quality tradeoff. NeST-Com exceeds adapted Ada in PSNR by 4.116/0.455/1.226 dB at 4/10/19 dB, respectively. At 10 dB, Ada has lower LPIPS and DreamSim and higher CLIP-I, with source-paired CIs excluding zero. Ada also has higher CLIP-I at 4 dB and slightly lower DreamSim at 19 dB; NeST has better PSNR, LPIPS, and CLIP-I at 19 dB.

Ada delivers all 256 tokens at 19 dB, so no tail sampling occurs; its approximately 58% source-capacity utilization reflects completion of its representation.

SwinJSCC and BPG have higher PSNR than NeST at every tested SNR. Latent JSCC has higher PSNR at 4/10 dB and lower LPIPS at 4 dB. Thus perceptual criteria and pixel error need not rank systems alike. NeST's PSNR advantage over ARPC is 3.26–3.97 dB. On matched100, replacing ARPC's original $\tau = 0.5$ sampling by argmax or $\tau = 0.25$ changes mean PSNR by up to about 0.15 dB in the tested conditions.

Fig. 3 shows two selected development examples. Ada retains crisp pastry folds and dog fur; NeST and Latent JSCC smooth different details, while ARPC changes visible object contours and texture. For the dog, Latent JSCC has the lowest displayed LPIPS, whereas Ada has the lowest DreamSim. These individual outcomes complement the source-level comparison.

On Kodak, all four paired metrics favor NeST over Ada at 4 dB. At 10 dB, only the PSNR difference has a 95% paired confidence interval excluding zero; at 19 dB, PSNR and LPIPS favor NeST while DreamSim/CLIP-I CIs cross zero.

D. Transmission With a Reduced Symbol Budget

At $N = 512$, 68 header symbols leave 444 body symbols. Policies are independently recalibrated on the calibration set using a common MCS pool. WHOLE achieves 3.79 and 3.66 dB higher mean PSNR than the fixed-frame Ada-TokenCom adaptation at 4 and 10 dB, respectively: a source-reconstruction advantage is already present with whole-scale endpoints. Within-scale transmission further improves PSNR over WHOLE by 0.39–0.79 dB across the three SNRs. It also improves LPIPS, DreamSim, and CLIP-I, with all source-paired 95% CIs excluding zero. At 10 dB, WHOLE and NeST-Com select the same target $(7, 0)$ and 16-QAM rate-2/3 MCS; within-scale fallback endpoints raise source-bit utilization from 79.30% to 93.46%.

TABLE III

REDUCED-BUDGET COMPARISON ON COMMON500 AT $N = 512$. $\Delta = \mathrm{NeST} - \mathrm{Ada}$; BRACKETS GIVE SOURCE-PAIRED 95% CIS. POSITIVE PSNR AND NEGATIVE DREAMSIM FAVOR NEST.

SNR	ΔPSNR (dB)	ΔDreamSim
4	$+4.180$	-0.1029
	$[+3.991, +4.377]$	$[-0.1142, -0.0917]$
10	$+4.055$	-0.0660
	$[+3.880, +4.235]$	$[-0.0747, -0.0573]$
19	$+1.740$	$+0.0139$
	$[+1.577, +1.909]$	$[+0.0103, +0.0173]$

Compared with Ada, NeST-Com has higher PSNR and lower LPIPS at every tested SNR, and lower DreamSim at 4/10 dB (Table III). Ada has higher CLIP-I throughout and lower DreamSim at 19 dB; these paired CIs exclude zero. The results support the next-scale representation and reconstruction system under a tight budget, with further gains from within-scale transmission. Cross-budget comparisons describe separately calibrated systems, whose candidate pools differ; they do not isolate representation order or a change of N alone. Body-CRC rejections are 4/7/5 for NeST/WHOLE/Ada at 4 dB, 1/0/0 at 10 dB, and zero at 19 dB, each out of 1500 trials. All remain in the means; no source requires NULL.

E. Additional Results and Processing Cost

The existing $N = 2048$ comparison uses 100 sources and two paid 1024-symbol subframes, with adaptive conditions at 4/10 dB and complete representations at 19 dB. At 19 dB, NeST exceeds ARPC by 1.806 dB PSNR, but ARPC has better DISTS [17]: NeST minus ARPC is $+0.0054\,[+0.0024, +0.0086]$.

With WHOLE/PARTIAL transmit policies fixed at 10 dB and RX using the true noise variance, all 300 receptions per method on a 100-image subset are rejected at 8 dB, whereas all succeed at 10/12 dB. For each source and method, the recovered prefix and reconstruction are identical at 10 and 12 dB; gray failures remain in the statistics.

Software processing latency at 10 dB averages 619.71 ms/image for NeST, 42.48 for Latent JSCC, 44.69 for SwinJSCC, 1520.30 for ARPC, and 2293.01 for BPG (RTX PRO 6000, six CPU cores, batch one, 16 development images measured three times after warm-up). Totals sum TX/RX software processing; Ada has not been timed under this protocol.

IV. Conclusion

NeST-Com combines within-scale endpoints, actual terminated-length fitting, and prefix-constrained reconstruction to adapt coarse-to-fine residual representations to fixed digital frames. Shared-model results show improved utilization and quality, with gains retained at 512 symbols. External comparisons support low-budget source reconstruction while revealing perceptual tradeoffs. Future work will address SNR uncertainty and fading channels.

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